





Mechanical Behaviour of Engineering Materials

↓
INTRODUCTION
to material

Groover: Chapters 2 and 5



The student:

- knows the nature of materials;
- knows the basic structure of metals;
- knows the basic structure of polymers.

*most important on
manufacturing
process*

*structure of
Most used in
manufacturing
process*





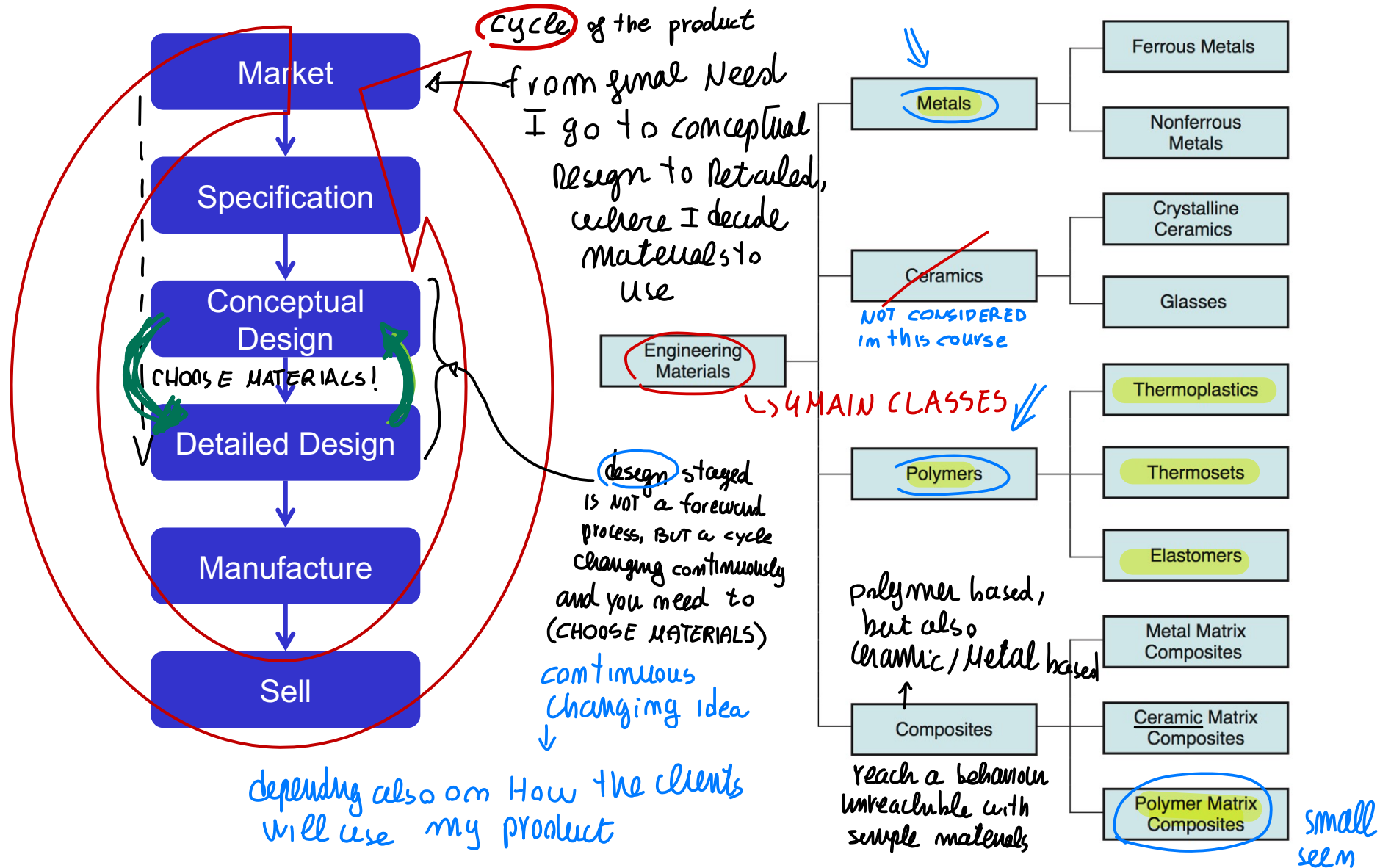
IMPORTANCE OF MATERIALS IN MANUFACTURING

Manufacturing is a transformation process:
↓ transforming materials

- It is the material that is transformed,
- And it is the behavior of the material when subjected to the forces, temperatures, and other parameters of the process that determines the success of the operation.



PRODUCT DESIGN AND MANUFACTURING



ELEMENT GROUPINGS

different kind of Metals...

5

The elements can be grouped into families and relationships established between and within the families by means of the Periodic Table.

Many chemical elements are metals!

1 IA	2 IIA											13 IIIA	14 IVA	15 VA	16 VIA	17 VIIA	18 VIIIA
1 H												5 B	6 C	7 N	8 O	9 F	10 Ne
2 Li	4 Be											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
3 Na	12 Mg	3 IIIV	4 IVB	5 VB	6 VIB	7 VIIB	8	9 VII	10	11 IB	12 IIB						
4 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
5 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
6 Cs	56 Ba	57-71	72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
7 Fr	88 Ra	89-103	104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Uut	114 Fl	115 Uup	116 Lv	117 Uus	118 Uuo
6	57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu	6	
7	89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr	7	

lot of metals in the material basics

Alkali Metals

Alkali Earth Metals

Transition Metals

Other Metals

■ **Metalloids**

Other Non Metals

 Halogens

 Noble Gases

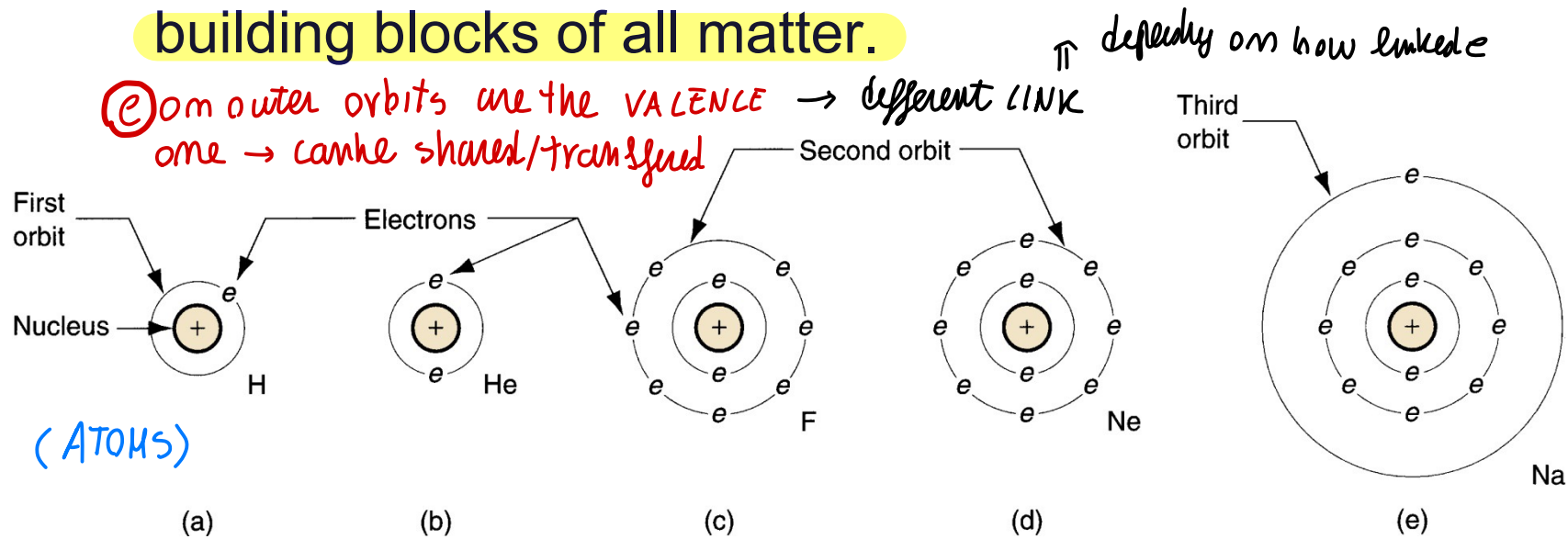
 Lanthanides & Actinides



ATOMIC STRUCTURE AND THE ELEMENTS

The basic structural unit of matter is the **atom**:

- Each atom is composed of a **positively charged nucleus**, **surrounded by** a sufficient number of **negatively charged electrons** so the **charges are balanced**;
- More than 100 elements, and they are the **chemical building blocks of all matter**.



(a) Hydrogen, (b) helium, (c) fluorine, (d) neon, and (e) sodium

depending on what happens at ATOMS level it changes the
material STRUCTURE → different materials



BONDING BETWEEN ATOMS AND MOLECULES

Atoms are held together in molecules by various types of bonds: *LINKS between elements*

1. **Primary bonds** - Characterized by strong atom-to-atom attractions that involve exchange of valence electrons.
Types: **Ionic**, **Covalent**, **Metallic**.
STRONGER Bonds

2. **Secondary bonds** - Generally associated with attraction between molecules, no transfer or sharing of electrons. Types: *Dipole forces*, *London forces*, *Hydrogen bonding*.

Primary bonds are much stronger than secondary bonds.

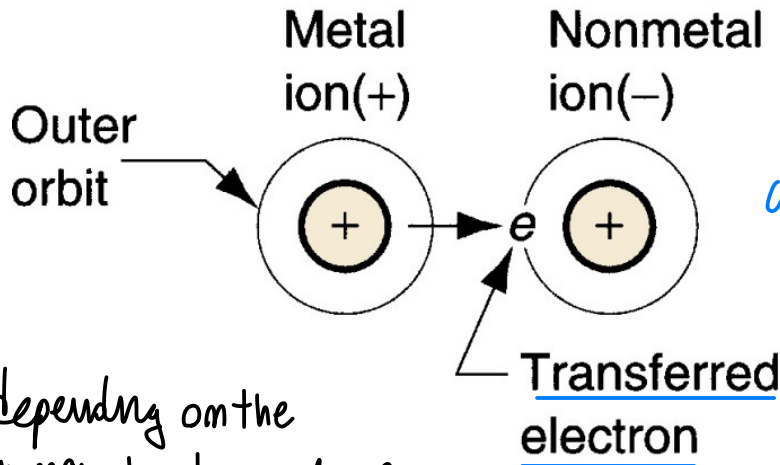


PRIMARY BONDS

8

Ionic Bond

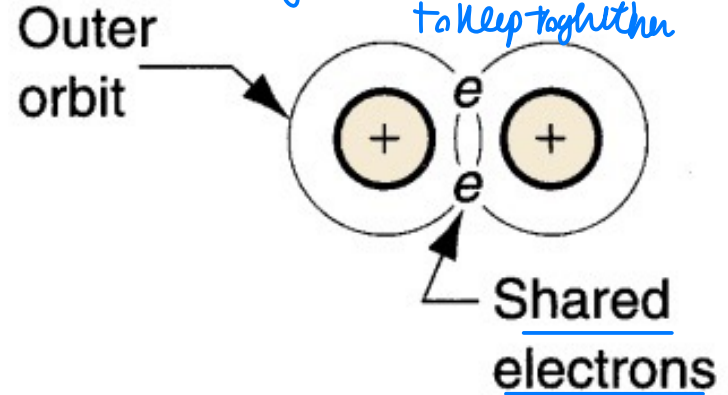
happens if valence electrons of external orbits of one chemical element is transf. to another
↓
creating (+) (-) ion



depending on the primary bonds material behaviours changes

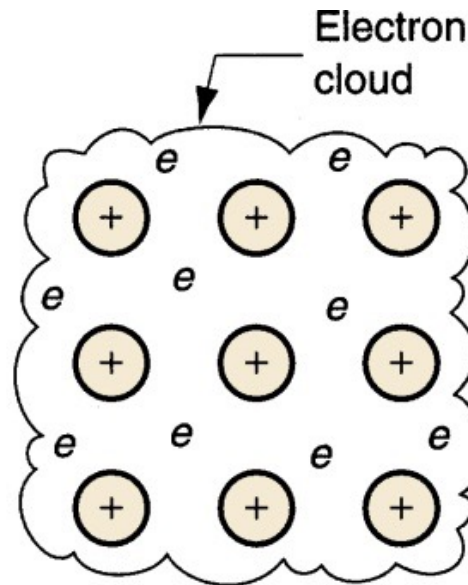
Covalent Bond

sharing the OUTER ORBIT to keep together



Metallic Bond

particular sharing of cloud of (e) (NOT the orbit) ⇒ share valence electrons all CLOUD!



cloud of sharing electrons

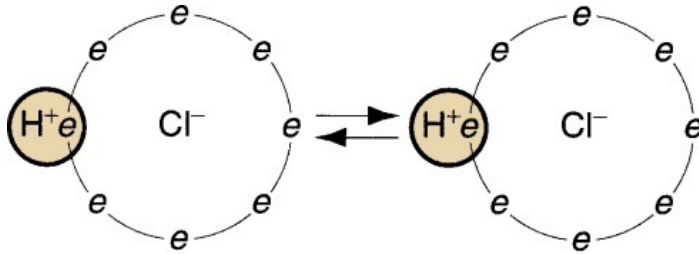
⇒ Change a LOT the Behaviours depending on PRIMARY BONDS!



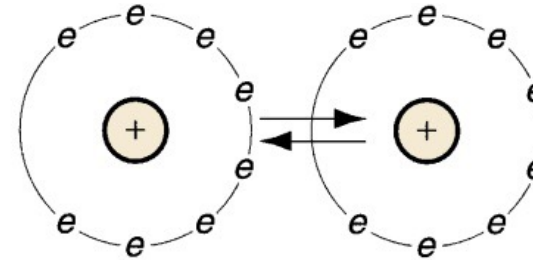
SECONDARY BONDS

creating a DIPOLE (molecule partition (+) / (-) to attract)

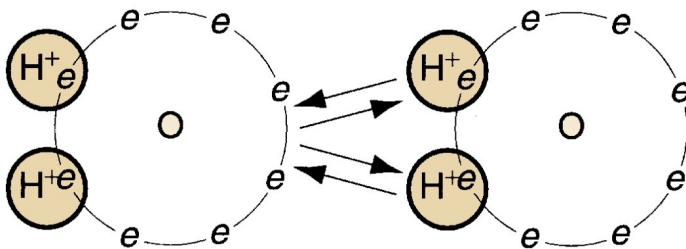
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Dipole Forces: Arise in a molecule comprised of two atoms with equal and opposite electrical charges. Each molecule therefore forms a dipole that attracts other molecules.



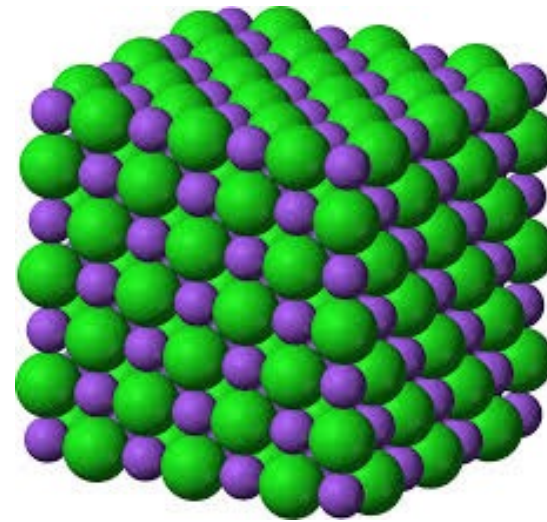
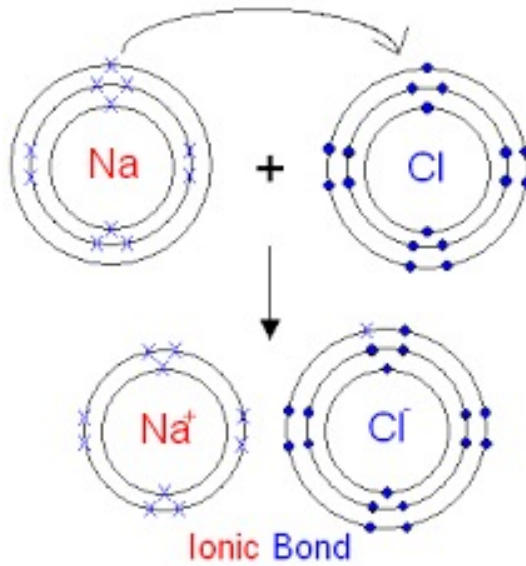
London Forces: Attractive force between non-polar molecules, i.e., atoms in molecule do not form dipoles. However, temporary dipoles form when more electrons are on one side.



Hydrogen Bonding: Occurs in molecules containing hydrogen atoms covalently bonded to another atom. Since electrons to complete shell of hydrogen atom are aligned on one side of nucleus, opposite side has a net positive charge that attracts electrons in other molecules.



EXAMPLE – CHLORIDE SODIUM





MACROSCOPIC STRUCTURES OF MATTER

STRUCTURE OF MATERIAL

Atoms and molecules are the building blocks of a more macroscopic structure of matter.

- When materials solidify from the molten state, they tend to close ranks and pack tightly, arranging themselves into one of two structures:

- **Crystalline**; → in which in a unit cell the atoms stays on a given ordinary position (denoted a priori) → repeating itself on the space
 - **Non-crystalline**. → Non ordinary structure (NONREGULAR)
- ↑
main structure of material



- **Metals:** ^{↑ thanks to metal bonds} crystalline structure in the solid state. High hardness, electrical and thermal conductivity, opaqueness, reflectivity.
- **Ceramics:** either crystalline or non-crystalline ^{↙ Both} structure (amorphous). High hardness, stiffness, brittleness, electrical insulation properties, refractoriness, chemical inertness.
- **Polymers:** ^{↑ basics of NON-CRYSTALLINE} either a glassy structure or mixture of glassy and crystalline structure. Low density, high electrical resistivity, low thermal conductivity. Strength and stiffness vary widely.
^{↑ a portion of polymers can (conserve) the crystal structure (macromolecule formed so that it similar to CRYSTALLINE)}



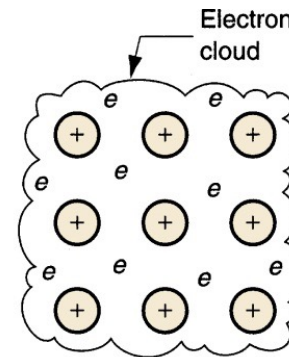
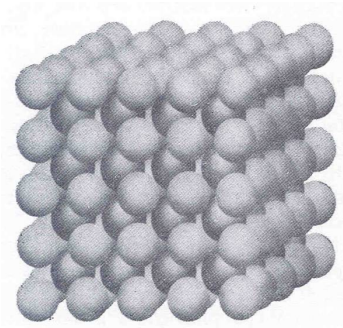
All metals, except for mercury, are in **solid phase at room temperature**. METALLIC BOND (PRIMARY)

At solid (and ideal) phase, **atoms are located in a regular structure (crystalline structure)**:

- Nodes $\Rightarrow$ Atoms;
- **Internal electrons close to nuclei;**
- **External electrons can move.**

*Basic cell repeat itself in space
showing electrons in
the cloud*

*With atom in
a fix position
on the space*



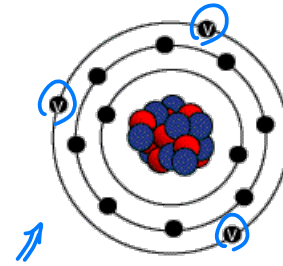


EXAMPLE – ALUMINUM

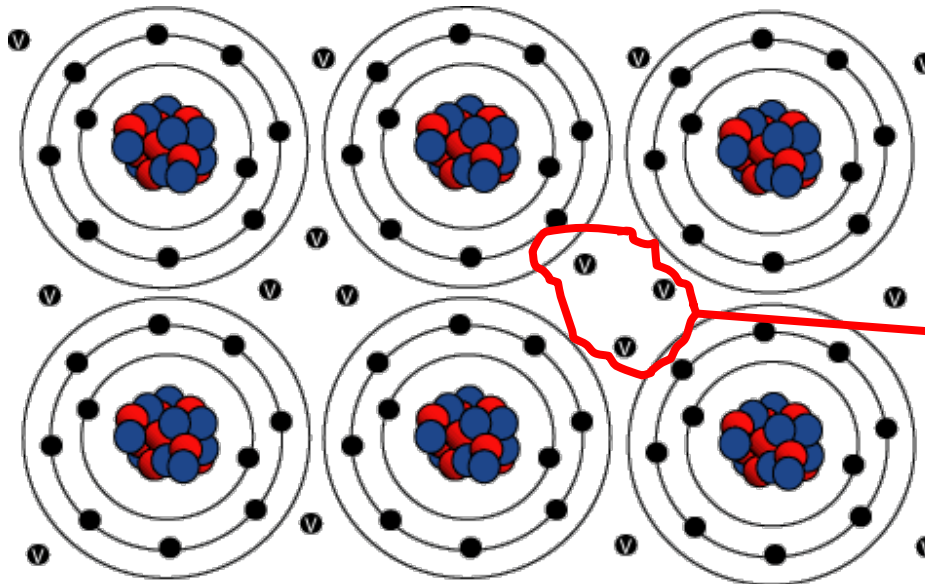
14

Aluminum atomic structure (13)

↓ OUTER ORBIT WITH
3 valence shared
electrons in CRYSTALLINE
STRUCTURE



3 valence electrons



Valence electrons
shared in the cloud.

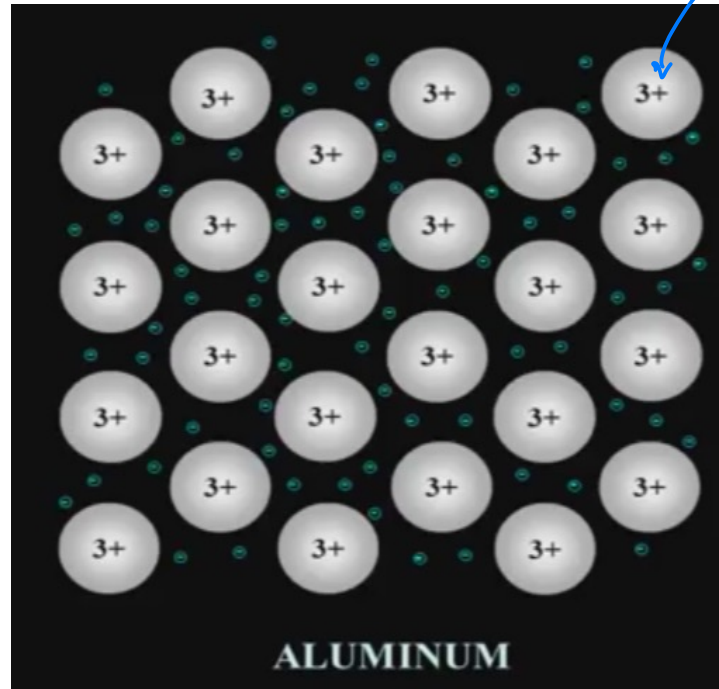


EXAMPLE – METALLIC BOND

how the
CLOUD
WORKS ↪

single ATOMS
(POSITIVES (+3))
↑
3 valence electrons
outside

3 balanced electrons

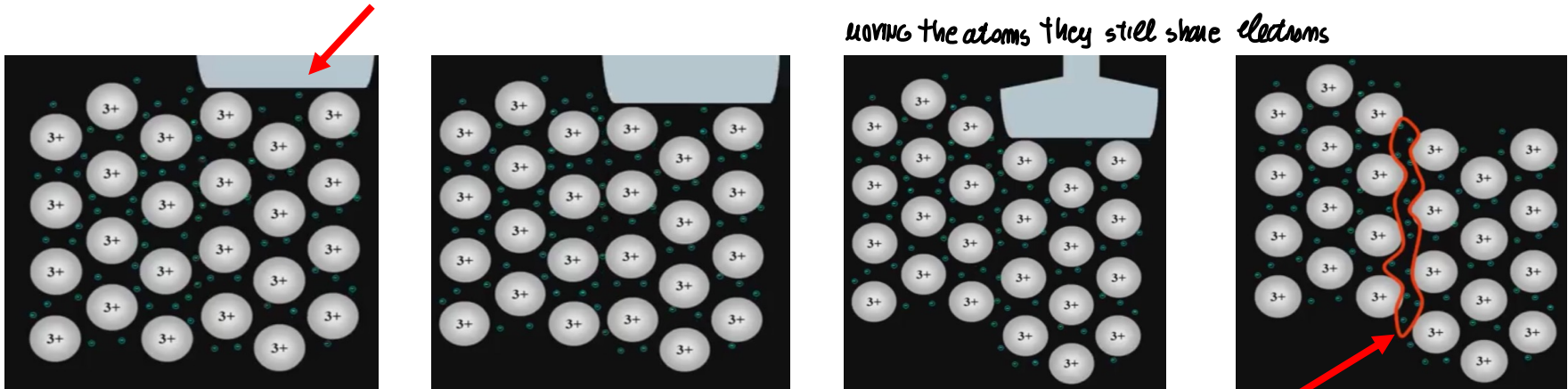


PROPERTIES FROM METALLIC BOND: MALLEABILITY

Applying a load to a portion of the structure

↑
tool

moving the atoms, they still share electrons
⇒ permanent deformation (still continuous)



MALLEABILITÀ

↓
proprietà di poter essere
deformato permanentemente
senza modifiche strutturali / fratture

The delocalized electrons
are a constant presence.

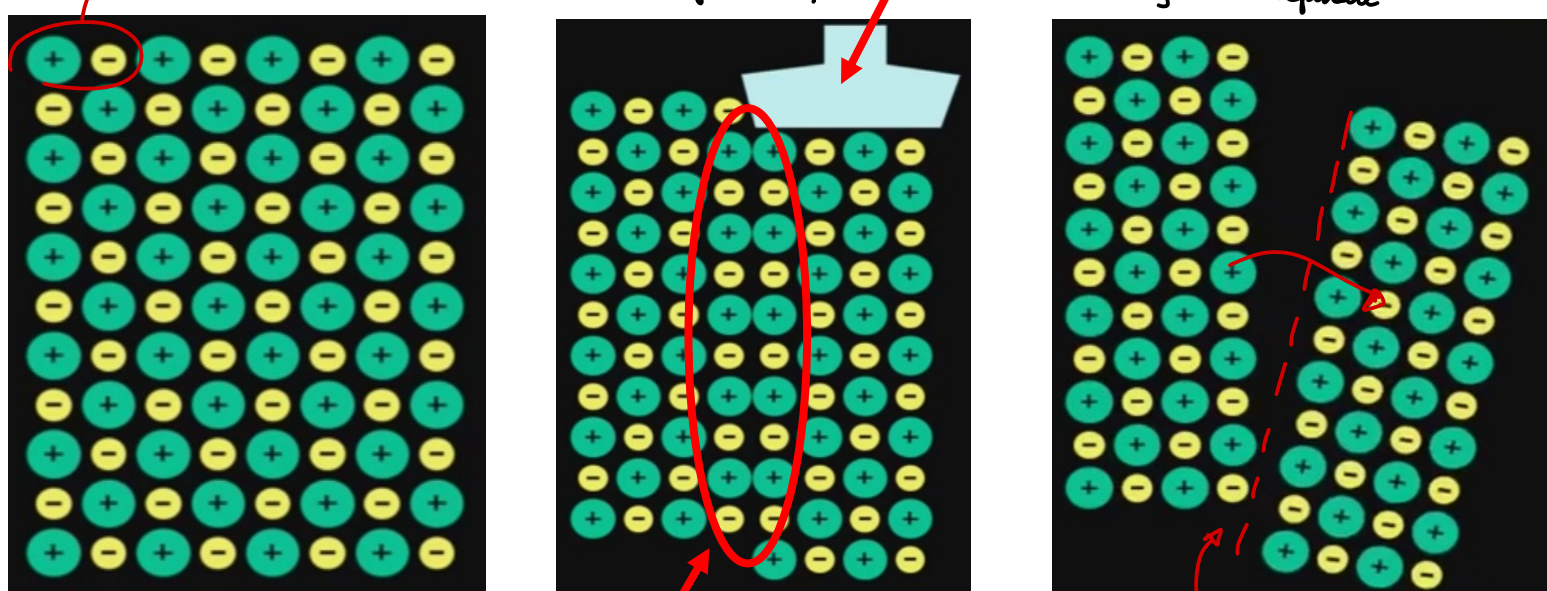


PROPERTIES FROM METALLIC BOND: MALLEABILITY

CERAMIC MATERIAL *valence went from one to another (+)(-) together*
Ionic structures are not malleable.

stay together

Load from outside, moving portion of structure
tool *so they could separate*
Moving I create a fracture! **CREATE A FRACTURE**



they separate
typical of brittle material, where no reformation but broken material

Charges are aligned
(like ceramic plate)



PROPERTIES FROM METALLIC BOND

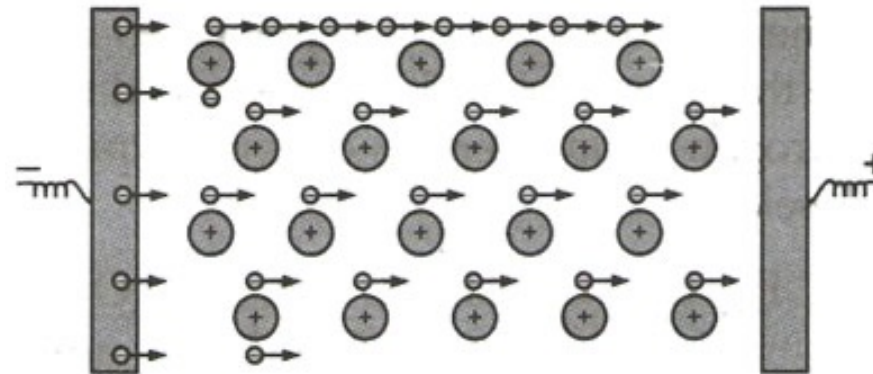
METALS

- Electrical conductivity: delocalized electrons also carry electronic current and heat because of their ability to move through crystals.

motion of electrons

electrical /
thermal
behaviours
due to the
CLOUD → valence
electrons can move
↓

transmission of heat and electricity



Electrical conductivity in metal crystals

→ propagation of HEAT
inside → thanks to motion
of electrons



PROPERTIES FROM METALLIC BOND

LIGHT Behaviours! (OPTICAL)

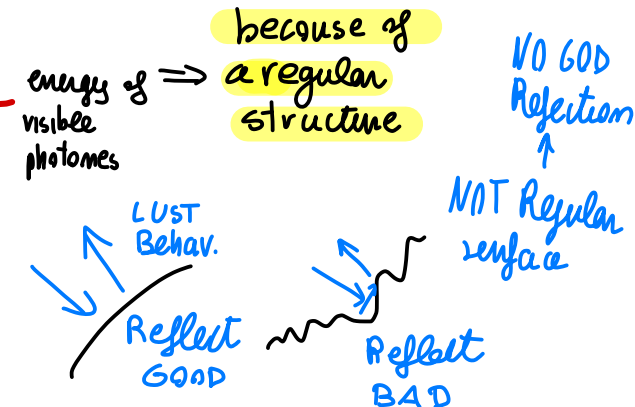
- **Lustre**: electrons readily absorb and re-emit visible frequency photons, giving metals their characteristic lustre. Surface roughness affects this property. What if you scratch?

electrons absorb energy and have LUST behave

↑ if high finishing surface
particular behaviour from
an OPTICAL point of view



surface
influence
final
behaviours



Cloud gate (A. Kapoor, 2006)
Millennium Park - Chicago
Stainless Steel

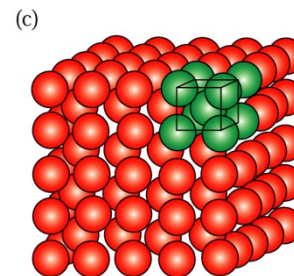
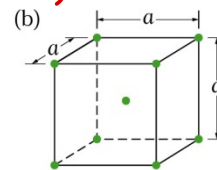
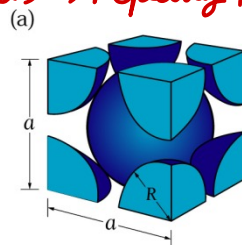
↳ always mirroring but if bad surface, work Bad

BASIC CRYSTAL STRUCTURES (UNIT CELLS) *→ Few examples*

↓ BASIC METAL CELLS (14 totals → repeating in space)

Body-Centered Cubic (BCC)

Metals: Iron (Fe- α , Fe- δ), Chromium (Cr), Molybdenum (Mo), Tungsten (W), ...

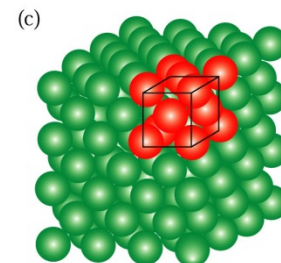
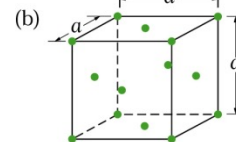
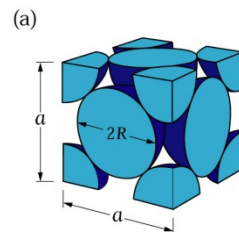


↑ IRON can take diff structure according to different values

DIFFERENT STRUCTURE for same chemical element

Face-Centered Cubic (FCC)

Metals: Iron (Fe- γ), Aluminium (Al), Copper (Cu), Nickel (Ni), Silver (Ag), Gold (Au), ...



difference in the position of the sphere

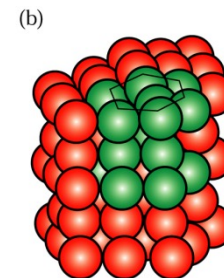
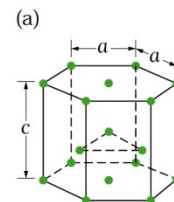
FACE CENTERED if in different position look next..

Hexagonal Close-Packed (HCP)

Metals: Magnesium (Mg), Titanium (Ti), Zinc (Zn), ... *different material TAKES different solution*

also same chemical elements can take different structures!

Iron: Fe- α , Fe- δ , Fe- γ . The appearance of more than one type of crystal structure is known as **allotropism** or **polymorphism** (meaning "many shapes").



HEXAGONAL is ABOVE and BELOW on same position

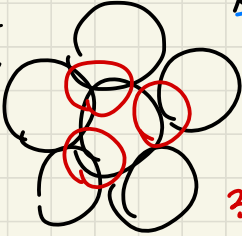
Atoms in Nature try to occupy space in the most efficient way

- Face-centered } most efficient way to occupy space
- Hexagonal } in nature $\rightarrow$ atoms occupy efficiently the space

- considering JUST one chemical element

Atoms $\approx$ sphere

Face-centered



most efficient
space occupation!

ON ONE
PLANE

$\rightarrow$ in 3D, the above
and below plane

3 upper plane

3 bottom

different position possible

same above / below HEX

different " FACE-CENTRIC



hexagonal



Different material $\Rightarrow$ Different solution!

IRON $\rightarrow$ near melting Fe- δ
midrange temp Fe- γ
room temp Fe- α } depends on temp
the STRUCTURE
of IRON

becoming stable in different temp range ---
with more material linked

MECHANICAL PROPERTIES OF MATERIALS

↓ Behaviour of material → MECHANICAL by loading the material in different ways to test it

Mechanical properties determine the behavior of a material when subjected to mechanical stresses.

Dilemma: mechanical properties desirable to the designer, such as high strength, usually make manufacturing more difficult.

- The manufacturing engineer should consider the design viewpoint.
- The designer should be aware of the manufacturing viewpoint.



MECHANICAL PROPERTIES OF MATERIALS

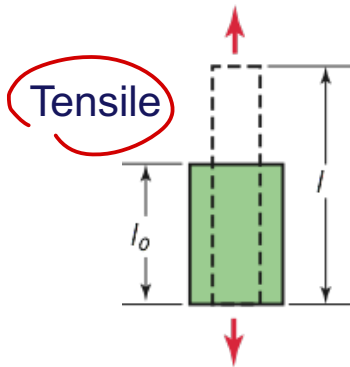
→ MECHANICAL LOAD to see properties BASIC LOAD: TENSILE LOAD

Three types of static stresses to which materials can be subjected:

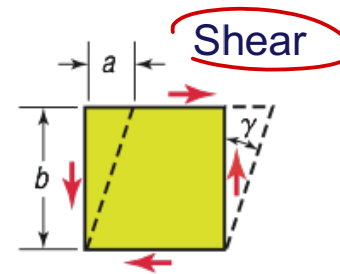
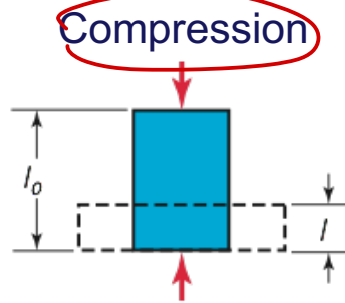
- Tensile - tend to stretch the material;
- Compressive - tend to squeeze it;
- Shear - tend to cause adjacent portions of material to slide against each other.

⇒ try to move adjacent portions

3 possible ways which we can load the material!



different loads, due to the STRESS I can have different behaviours



→ how they go together in different situations

Stress - Strain curve: basic relationship that describes mechanical properties for all three types.

from LOAD → we can use TESTS to see the behaviours (PROPERTIES)

⇒ STRESS → Deformation STRAIN



IMPERFECTIONS

IMPORTANT... considering a CRYSTAL structure
of pure metals → performance higher respect
to real materials

23

real material has defects respect to ideal CRYSTAL

- In theory metals should resist more than they do in reality.
for the crystal structure
↓ but IN REALITY we have
- The presence of imperfections reduces the resistance of
metals: on the material
 - Point defects: interstitial atom, vacancy, impurity atom;
 - Linear defects: dislocations;
 - Surface defects: grain boundaries, phase boundaries;
 - Volume defects: voids, inclusions, other phases, cracks.

Metallic bond give us

Some properties → mechanically due
to defects we have different behaviours!



POINT DEFECTS (Imperfection)

from CRYSTALLINE structure → small defects

24

Small defects
influencing the
behaviours
↓
less performances!

(INTERSTITIAL POSITION)

Self-interstitial

atom

(*)

if some atom
is missing,
someone is
stretched, in
a place where it
is NOT expected!

Vacancy

we expect an atom-
missing
↑

MISSING (MISSING NODES)
ATOMS in the structure

BASIC
ELEMENTS

DEFECTS

different chemical
element in the structure

Substitutional

impurity atom

↑
another
chemical
element
↓

deform structure
around
with similar dimension...
DISCONTINUITY

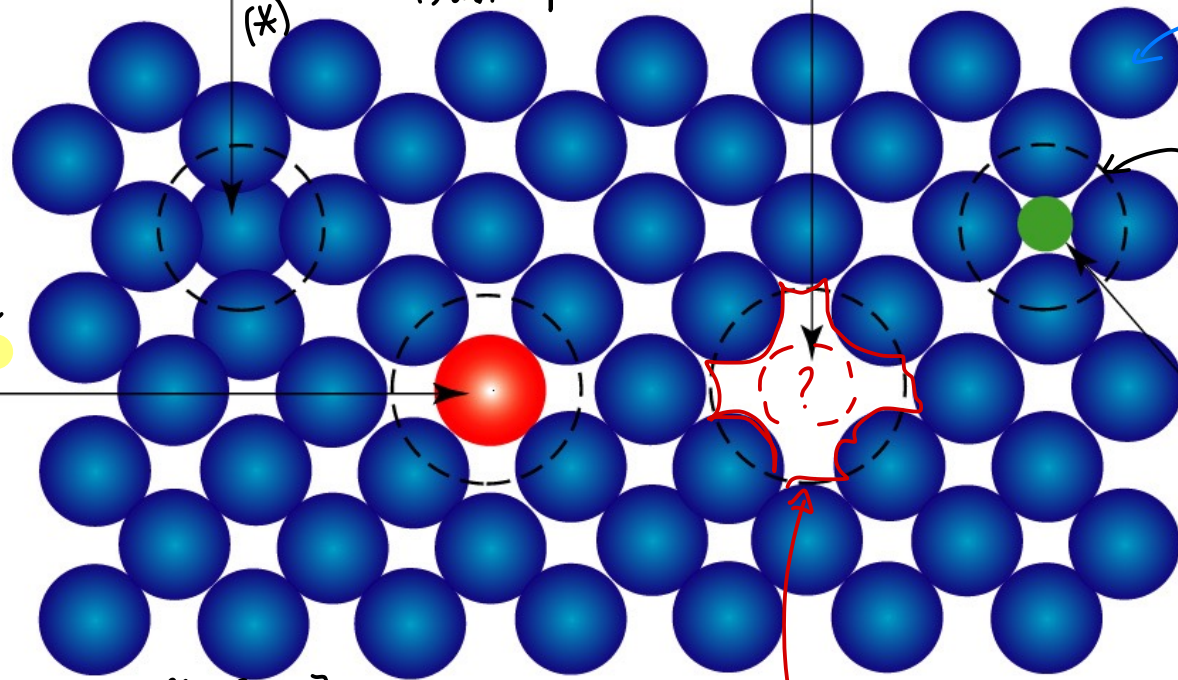
[usually low
energy, good
positioning!]

if ATOM
MISSING, appears
(*) When an ATOM
is in a place where NOT EXPECTED

on this boundary
we have a DISTORTION
of atom position ⇒ so energy inside,
reducing properties

same
element
Interstitial
impurity atom
in an interstitial
position →

deforming close
structure with
Impure Atom



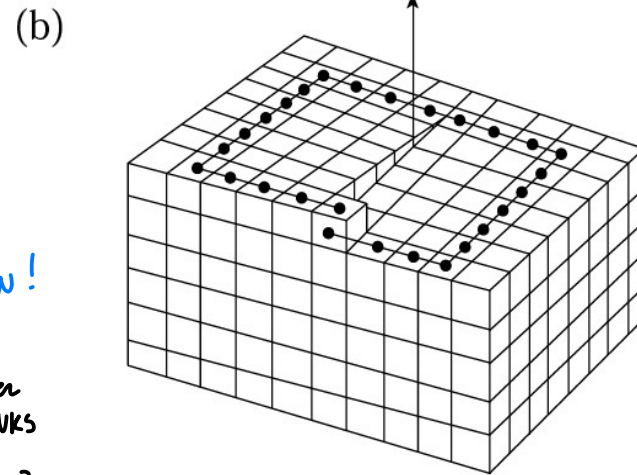
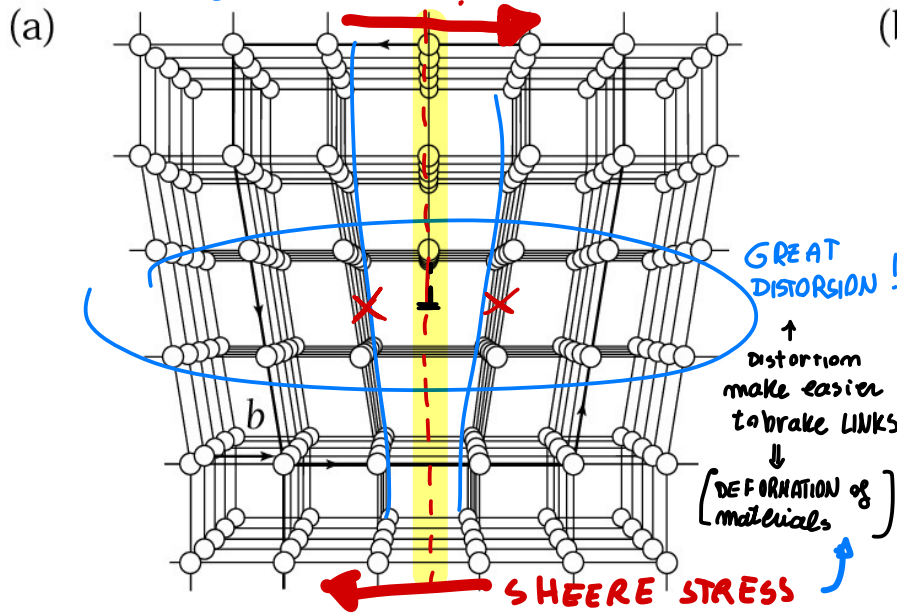


LINEAR DEFECTS

BIGGER Defects THAN VOLUME Defects

25

deforming the nearby plan ← partially missing
UNCOMPLETE PLAN Vertically!



Edge dislocation

(DISTORSION)

A linear defect at the edge of an extra plane of atoms.

Vertical PLAN --

When loading when shear stress -- ⇒

create huge deformation of the STRUCTURE

loading, we can remove boundaries (X) and create new ones

↓ easier LINK BREAK.

Screw dislocation

A helical defect in a three-dimensional lattice of atoms. Screw dislocations are so named because the atomic planes form a spiral ramp.



ELASTIC STRAIN

recover initial condition

26

ELASTIC BEHAV.

ELASTIC BEHAV.

STRESS that does NOT modify CRYST STRUCTURE
ELASTIC RECOVERY

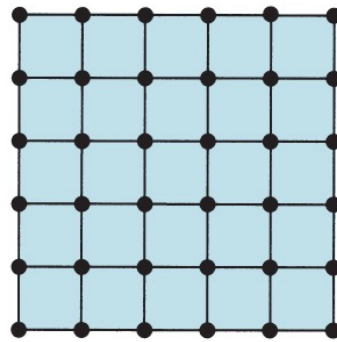
When a crystal experiences a gradually increasing stress, it first deforms elastically.

bring the stress to fracture the structure

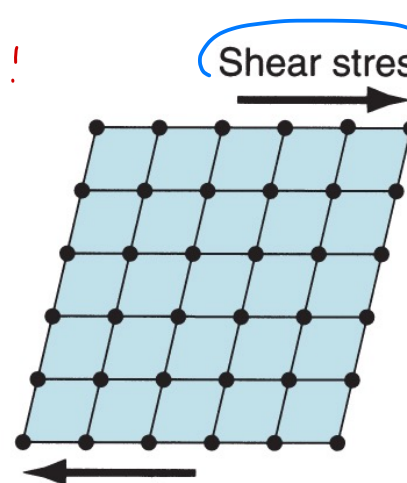
Removing the LOAD the structure create a permanent deformation ← NOT here!

If the LOAD is enough high (IN SLIDE 27)

↓
While here, With less load, it came back to original pos.



(a)



(b)

Shear stress

STRESS so that some LINK are fractured

→ permanent def

ELASTIC behaviour of a METAL

Deformation of a crystal structure: (a) original lattice: (b) elastic deformation, no permanent change in positions of atoms.

PLASTIC STRAIN

Breaking links applying a load
creating new links

EDGE DISLOCATION is a
useful properties → permanent DEF.

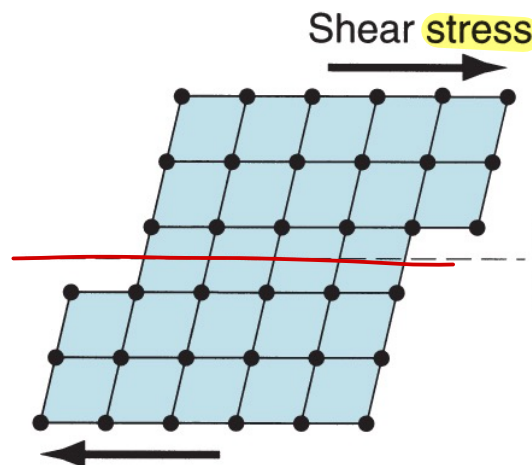
↳ permanent Deform

If the stress is higher than forces holding atoms in their lattice positions, then a permanent shape change occurs.

FINAL
LOAD
RESULT

↓
removing the
load - atoms
remain on
that position!

NO GO BACK @ original position (c)



different from
elastic behaviour that
turn to the initial
condition...

recover elastically
but NOT to initial
condition

for emean defects We
destroy bonds creating
new one ⇒ permanent deformation

Plastic deformation (slip), in which atoms in the crystal lattice structure are forced to move to new "homes".

removing STRESS..

it comes back to a different position!

DEFORMATION IN METALLIC CRYSTALS

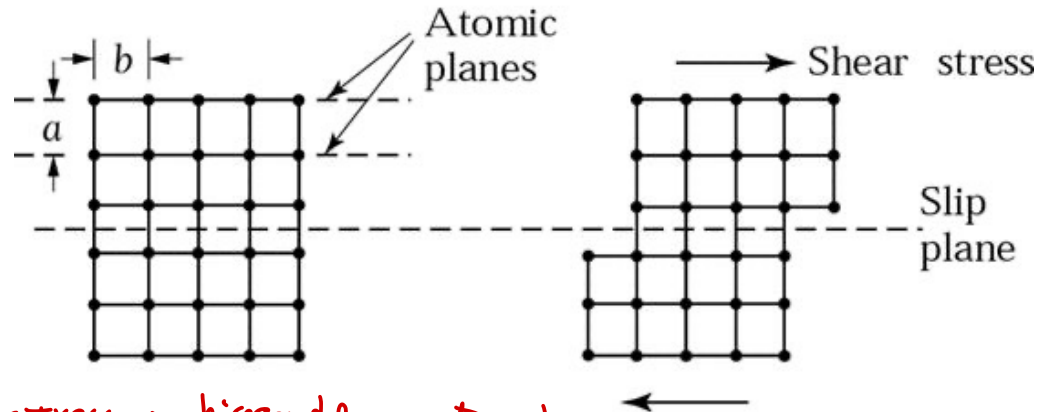
↓ In case of a DISLOCATION (discontinuity)

Deformation of a crystal structure

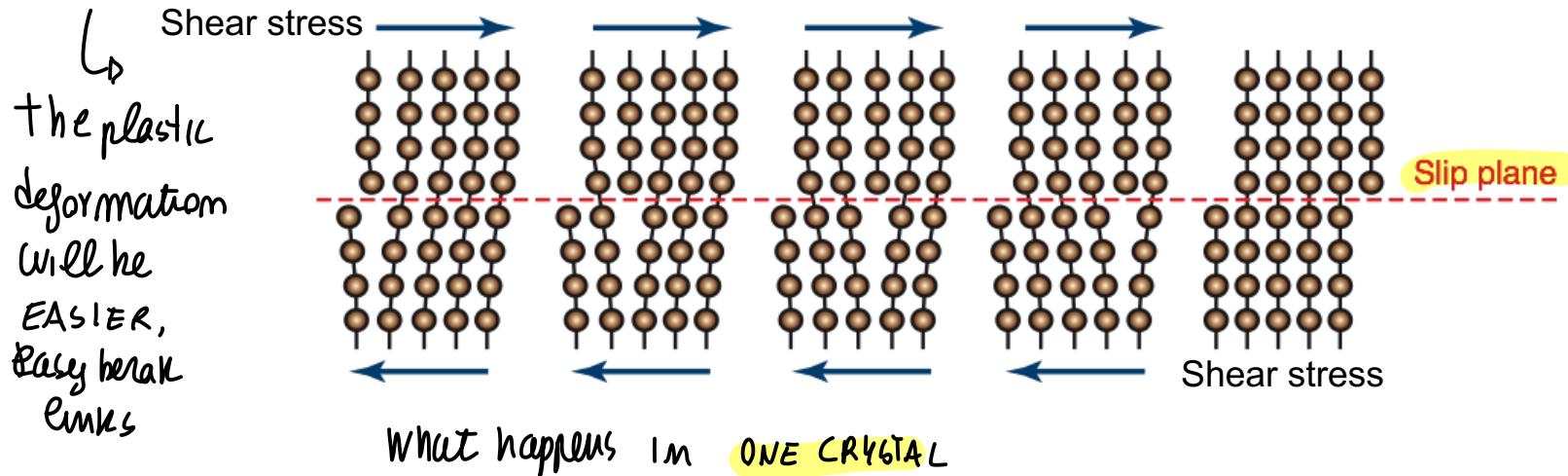
DISLOCATION that eases the plastic deviation ...

With a LINEAR DEFECT the deformation is helped

→ less stress ⇒ bigger deformation!



Effect of dislocations in the lattice structure under stress.





POLYCRYSTALLINE NATURE OF METALS

↑ NOT just one crystal...

A block of metal may contain millions of individual crystals, called grains. Such a structure is called polycrystalline: structure (*)

- Each grain has its own unique lattice orientation;
- But collectively, the grains are randomly oriented in the block.

Structure is NOT JUST ONE CRYSTAL ⇒ usually classical material has (*)

How do polycrystalline structures form? ↗ due to SOLIDIFICATION PROCESS ⇒

- As a volume of metal cools from the molten state and begins to solidify, individual crystals nucleate at random positions and orientations throughout the liquid;
- These crystals grow and finally interfere with each other, forming at their interface a surface defect - a grain boundary, which is a transition zone, perhaps only a few atoms thick.



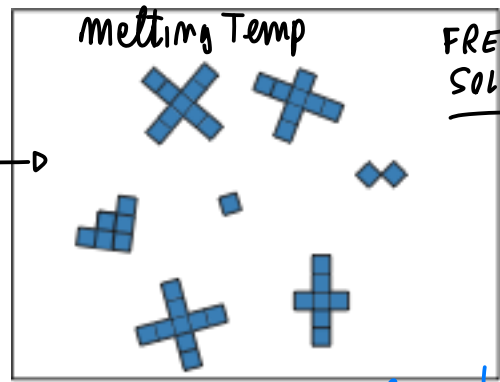
to be understand, you need to understand SOLIDIFICATION

SURFACE DEFECTS: GRAINS

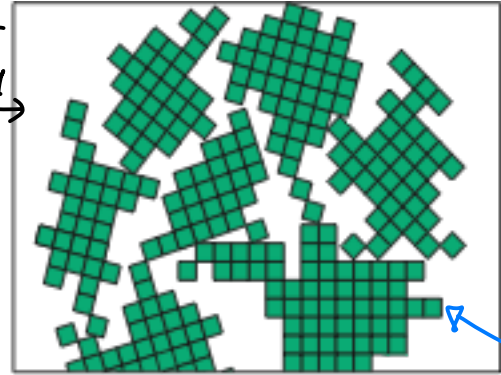
polycrystalline structure

solidification starts... obtaining random crystal

(white Area)
LIQUID MATERIAL



FREEZE
SOLIDIFY



Inside liquid, the ATOMS take position inside the NODES on the structure

Nucleus of CRYSTALS create randomly, depend to the removing heat (COOLING RATE)

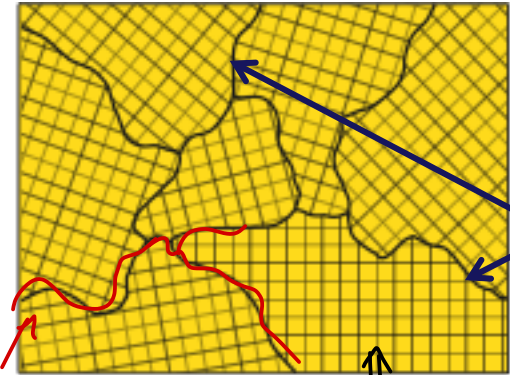
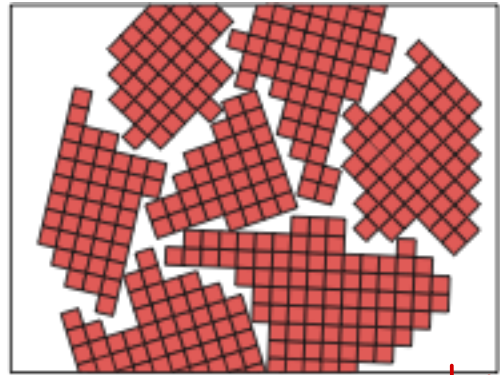
(a)

(b)

Nucleum depends on COOLING RATE

(removing heat, reach a situation where you have less boundaries)

remove heat
less energy
equilibrium better than liquid



(CRYSTAL) Grain boundaries different orientations

AT the end we have polycrystalline

interface between GRAIN

COMPLETE SOLIDIF.

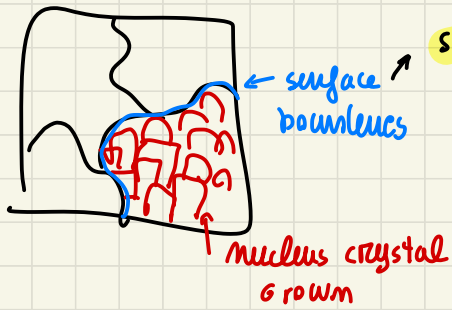
ANY ATOM free to add to the crystal

ATOMS NOT part of nucleus, so IN NO GRAIN

Various stages during solidification of molten metal.

Surface defects

REMOVING HEAT! GROWING of CRYST STRUCT. nucleus
Different nucleus grow with different direction



surface among different
GRAIN ARE
defects of our material!



GRAINS

3D elements CRYSTALS → surface boundaries of GRAIN ⇒ surface Defects
(mix of different elements) ALLOY

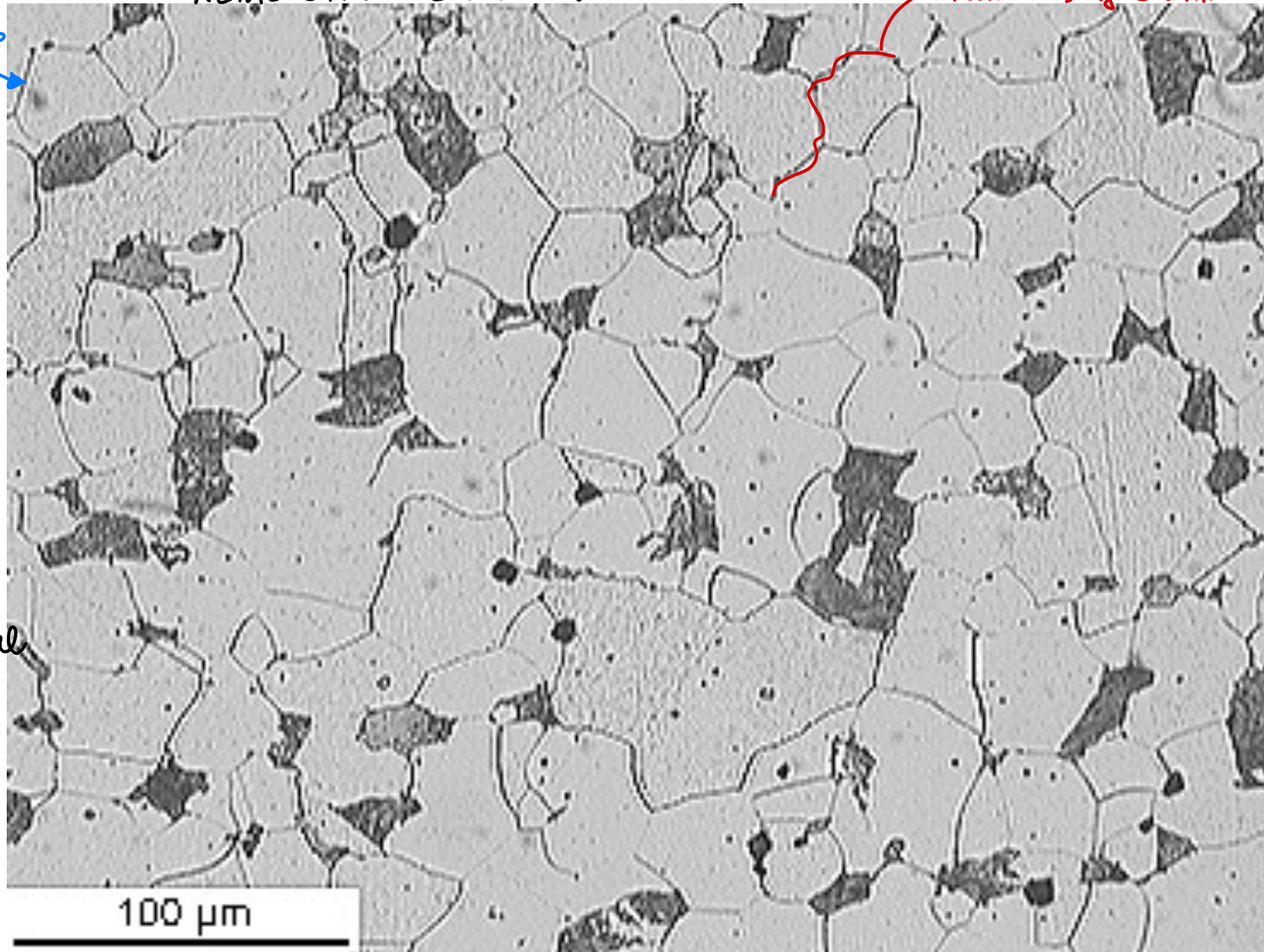
31

METAL CROSS SECTION

Boundaries of GRAIN

Boundaries create breaks in the material

POLYCRYSTAL Metal structure
↓
with ALLOYS
↓
mix of chemical elements





METAL (ALLOYS)

→ mixing of different chemical elements
(come of them METAL)

32

You can have more elements

Alloy: composition of two or more chemical elements, at least one of which is metal.

Two basic forms:

- **Solid solutions:** A solute (minor element) is added to the solvent (major element). Substitutional and interstitial solid solutions are possible.

REFERENCE

define crystal structure

↑ INSIDE the SOLVENT

↑ inside the structure

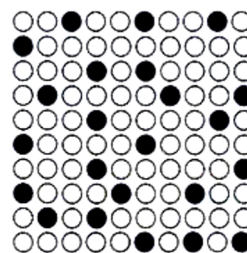
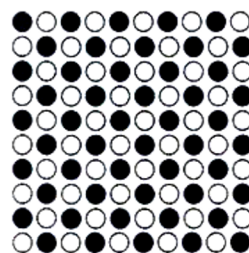
↑ atoms of different chemical element

- **Intermetallic compounds:** Complex structures consisting of two metals in which solute atoms are present among solvent atoms in certain proportions.

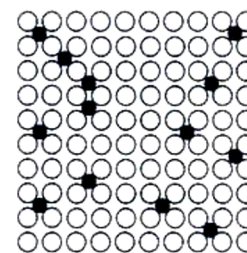
if MINOR > MAJOR, we have interstitial space

equivalent ↑ proportion of the solution parts !

Substitutional Solid Solution



Random Substitutional Solid Solution



Interstitial Solid Solution

Different structure = different behaviour

↑

I can add chemical element and
make a mix to change the behaviours

↓

like IRON that has diff. behav. with diff. structure, to work
far away from room temp

↳ we change behav. of material!

Solvent. characterize crystalline structure of Alloy

substitutional
solid solution

⇒

Atoms wanted take the
place of some moles inside CRYSTALLINE
↓ structure (well proportion / Random)

- if minor is smaller than solvent ⇒ INTERSTITIAL solid solution
(so the Atoms take an INTERSTITIAL SPACE)
- if exact proportion of elements ⇒ INTERMETALLIC compounds



POLYMERS

33

[Main characteristics of material...]

elements needed to understand behavior of materials
and transform them → manufacturing

Defined as a compound consisting of long-chain molecules, each molecule made up of repeating units connected together.

- The word *polymer* is derived from the Greek words *poly*, meaning many, and *meros* (reduced to *mer*), meaning part. ("Many parts") → macromolecule with a repeated unit along the space
- There may be thousands, even millions of units in a single polymer molecule.
- Most polymers are based on carbon and are therefore considered organic chemicals.

macromolecule

↳ (unit repeated along space)



long chain of repeating unit



TYPES OF POLYMERS

↓ classification:

Polymers can be separated into **plastics** and **rubbers**.

As **engineering materials**, it is appropriate to divide them into the following three **categories**:

- 1. Thermoplastic polymers;
 - 2. Thermosetting polymers;
 - 3. Elastomers;
- } PLASTICS

3. Elastomers; → Rubber .. With particular behav: they can change their shape applying a load, BUT it is an elastic elongation!

where (1) and (2) are plastics and (3) are rubbers.

Well elongation of the polymer

↑
return to the initial condition

↑
behaves between thermoplastic/thermoset

↓
go back to initial condition



THERMOPLASTIC POLYMERS OR THERMOPLASTICS

↓
material that can be shaped many times (using heat)

Solid materials at room temperature, but viscous liquids when heated to temperatures of only a few hundred degrees:

MAIN PROCESS to have this element is
chemical reaction $\leadsto$ Reshapeable

- This characteristic allows them to be easily and economically shaped into products;
- They can be subjected to heating and cooling cycles repeatedly without significant degradation;
- Symbolized by TP.

MANY TIMES!

liquify the material (high viscosity)

and shape the material



↓ Freeze and come back to solid

using temperature $\rightarrow$ melting to shape multiple time the material
(chain of elements thanks to a reversible chemical reaction)

THERMOSETTING POLYMERS OR THERMOSETS

↓ here solidification process (POLYMERIZATION) IS NOT REVERSABLE (only once)

Cannot tolerate repeated heating cycles as thermoplastics can:

↓ re-heating
the object brings
to Burning

- When initially heated, they soften and flow for molding (pre-polymers);
- Elevated temperatures also produce a chemical irreversible reaction that hardens the material into an infusible solid;
- If reheated, thermosets degrade, burn or char rather than soften;
- Symbolized by TS.

↑ "carbonizzare"

pre-polymer (liquid) → shape it by → solidification process
giving energy



ELASTOMERS (RUBBERS)

Polymers that exhibit extreme elastic extensibility when subjected to relatively low mechanical stress:

- Some elastomers can be stretched by a factor of 10 and yet completely recover to their original shape;
- Although their properties are quite different from thermosets, they share a similar molecular structure that is different from the thermoplastics.



MARKET SHARES ||

(70%)

38
↓ RECYCLING PLASTIC

Usage of this materials... MOSTLY are Thermoplastic material that can be recycled

Thermoplastics are commercially the most important of the three types: $\triangleright$ remelted, reshaped $\rightarrow$ melt + resolidify with small problem

- About 70% in weight of all synthetic polymers produced;
- Thermosets and elastomers share the remaining 30% about evenly, with a slight edge for the former.

[PROBLEM]

$\uparrow$ RECYCLING ISSUES! NOT Reshapeable, NOT Reversible solidification
can't be melt

On a volumetric basis, current annual usage of polymers exceeds that of metals.

(+ more object to give color)

IF using additional element
(like colorant chemical)

$\downarrow$ brings to degradation
if melted



REASONS WHY POLYMERS ARE IMPORTANT

MORE volumes of plastic than metals $\rightarrow$ competitive on shape material

Plastics can be molded into intricate part shapes, usually with no further processing: *processes of metal / plastic shape*

- Very compatible with *net shape* processing.

On a volumetric basis, polymers:

- Are cost competitive with metals;
- Generally require less energy to produce than metals.

Mechanical behav. issues $\rightarrow$ *polymers can't be the base, mainly for composite material (better mech. critically)*

Certain plastics are transparent, which makes them competitive with glass in some applications.

Respect to metals $\Rightarrow$ we have more plastic volumes than metals ones

Less shaping cost $\rightarrow$ energy needed to shape polymers $\ll$ metals

Net-shape process $\Rightarrow$ from raw material $\rightarrow$ just one step to obtain final product



SYNTHESIS OF POLYMERS

from a starting point (monomer)
(unit) starting element → chemical reaction
↑
to put together

40

↓ chemical reaction to have solid Polymers: **POLYMERIZATION**

Nearly all polymers used in engineering are synthetic:

- They are made by chemical processing.

Polymers are synthesized by joining many small molecules together into very large molecules, called macromolecules, that possess a chain-like structure.

The small units, called *monomers*, are generally simple unsaturated organic molecules, such as ethylene C_2H_4 .

↳ two kind of reactions...
 { chain reaction
 { adding macromolecular for each step



EXAMPLE POLYETHYLENE

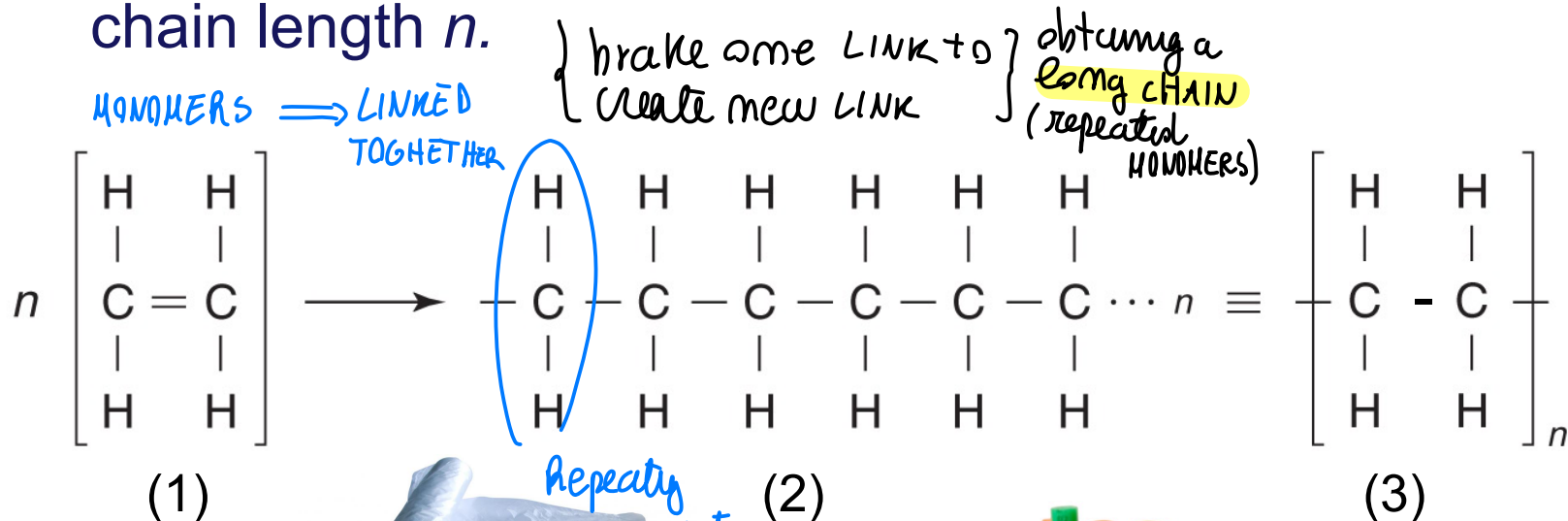
chemical reaction should bring to the separation...
repeating chemical unit

41

↓ STRUCTURE of POLYMERS

Synthesis of polyethylene from ethylene monomers:

- 1) n ethylene monomers, (from monomers)
- 2) polyethylene of chain length n ; *from chemical Reaction (separation of link to link others)*
- 3) concise notation for depicting polymer structure of chain length n .



repeating unit

High number of repeated monomers





POLYSTYRENE ("POLISTIRILO")

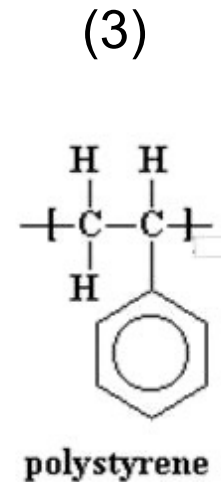
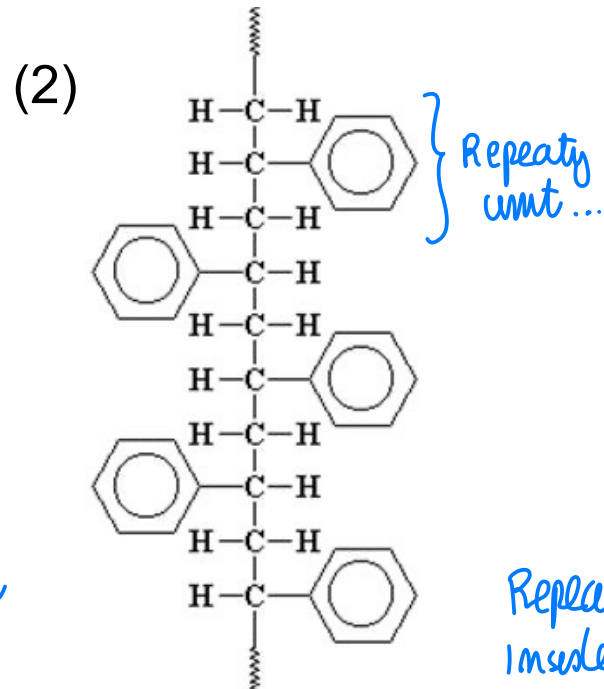
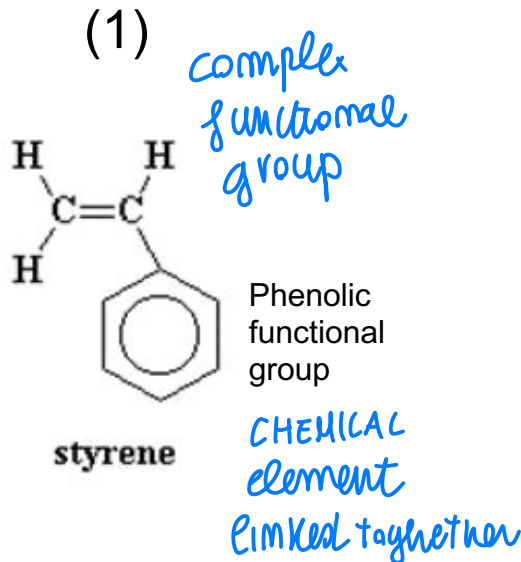
Just changing one ATOM/LINK
I create a total New Material

42

Synthesis of polystyrene from styrene monomers:

- 1) n styrene monomers, from a monomer...
- 2) polystyrene of chain length n ;
- 3) concise notation for depicting polymer structure of chain length n .

POLYMERIZATION:
break one LINK between two carbon atoms and make new linkage creating long chain



Repeat unit inside macromolecular





SYNTHESIS OF POLYMERS

Monomers can be linked into polymers in repeating units to make longer and larger molecules by a chemical process called a **polymerization reaction**.

Although there are several variations, two polymerization processes are important:

↓ 2 KIND of REACTION..

1. condensation polymerization;
2. addition polymerization.



In condensation polymerization, polymers are produced by the formation of bonds between two types of reacting mers.

A characteristic of this reaction is that *reaction by-products* (such as water H_2O , or hydrochloric acid HCl) are condensed out.

This process is also known as *step-growth* or *step-reaction polymerization*, because the polymer molecule grows step-by-step until all of one reactant is consumed.



SYNTHESIS OF POLYMERS

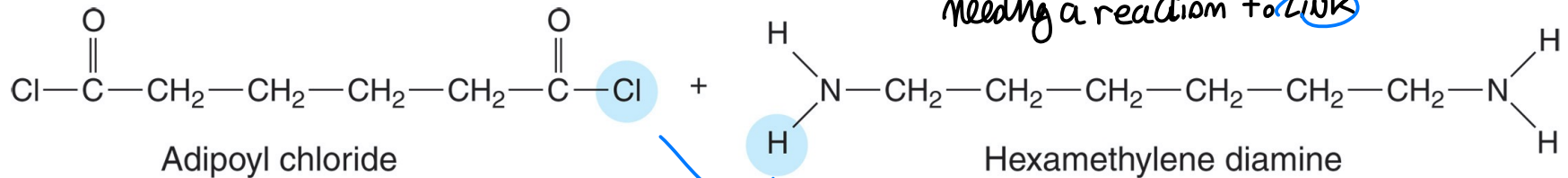
45

Reaction which link together the assets... create a LONG CHAIN
in the macromolecule

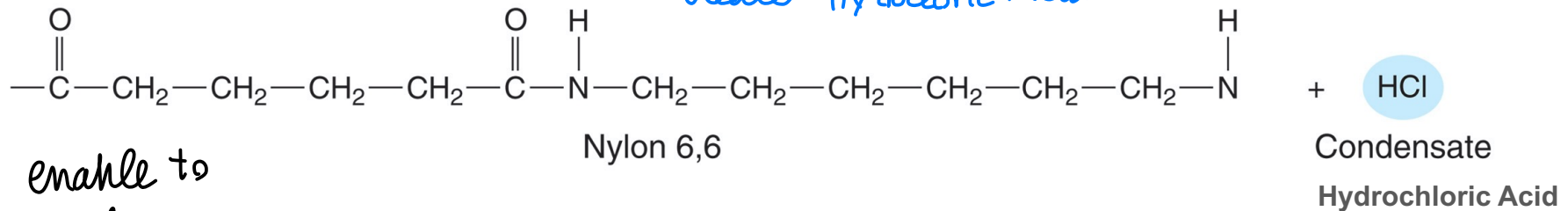
two atoms H, Cl

Condensation polymerization of nylon 6,6

starting from simple monomers → after **reaction**... you obtain **Macromolecule**
needing a reaction to **LINK**



LINK together to
create Hydrochloric Acid → going Away



enable to
create
LONG
chain
↓
adding chain
to chain





In addition polymerization (also called *chain-growth* or *chain-reaction polymerization*), bonding takes place without reaction by-products.

It is called “chain reaction” because of the high rate at which long molecules form simultaneously, usually within a few seconds.

This rate is much higher than that in condensation polymerization. In addition polymerization, an *initiator* is added to open the double bond between two carbon atoms, which begins the linking process by adding many more monomers to a growing chain.

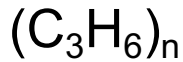


SYNTHESIS OF POLYMERS

Addition polymerization

like a chain reaction I add part by part
the elements

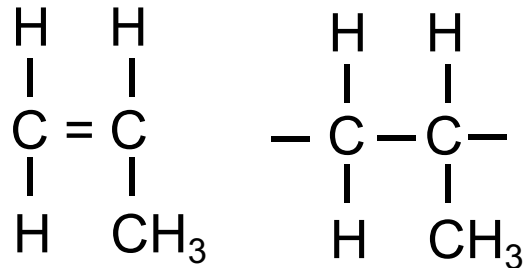
Polypropylene (PP)



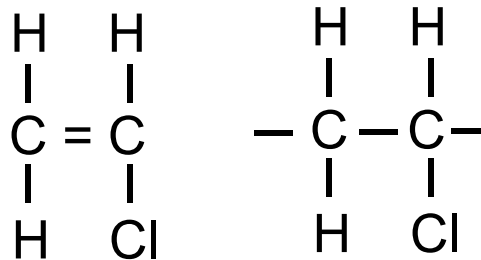
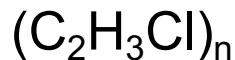
1) open a link on one monomers

2) enable to create a new link

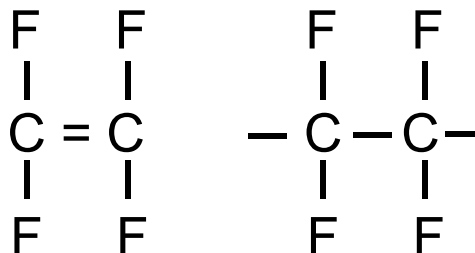
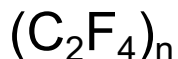
→ chain reaction, without any losing product..



Polyvinyl chloride (PVC)



Polytetrafluoroethylene
(PTFE - Teflon)





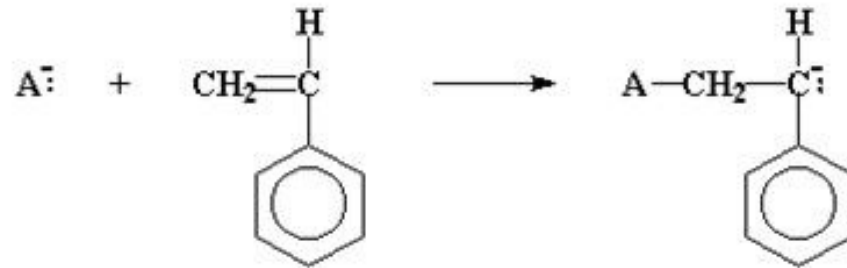
SYNTHESIS OF POLYMERS

$\Delta t_{\text{CONDENSATION}} > \Delta t_{\text{ADDITION}}$
polymerize
faster

48

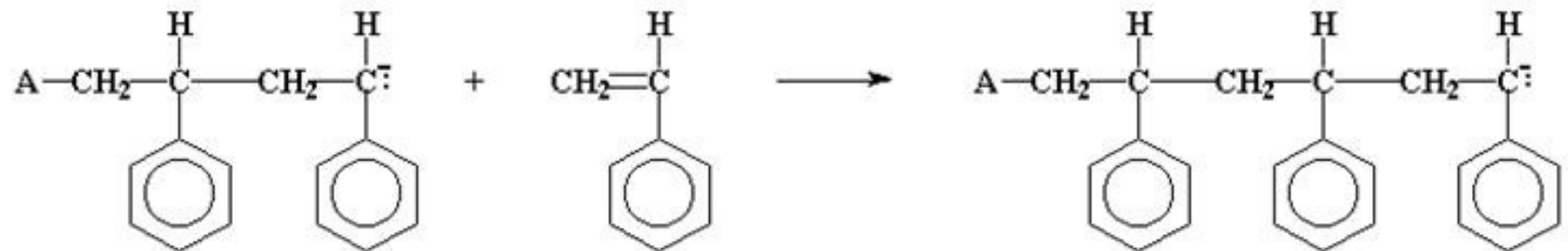
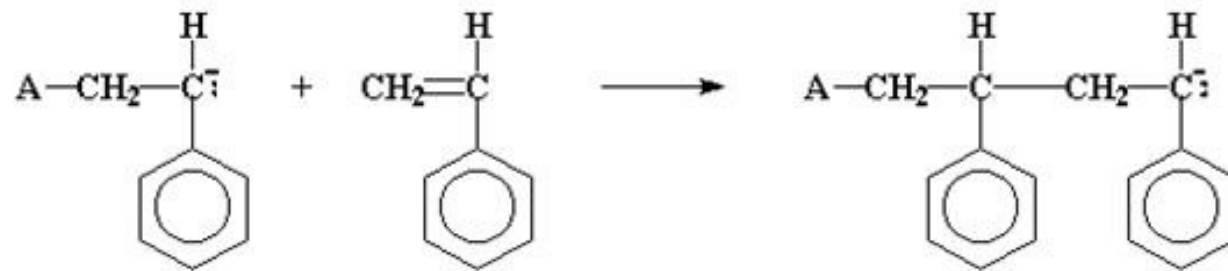
↙ faster reaction than the other! → important in the shaping point of view

Anionic addition polymerization of styrene



to start
We need

ADDITION polymerization
BICIRCLE ⇒ elements able to link atoms and create chain



Addition polymerization FASTER than CONDENSATION's one



important on SHAPING! $\Rightarrow$ impact on shape P.O.V

Waiting on polymerization I know

solidification is faster if Addition

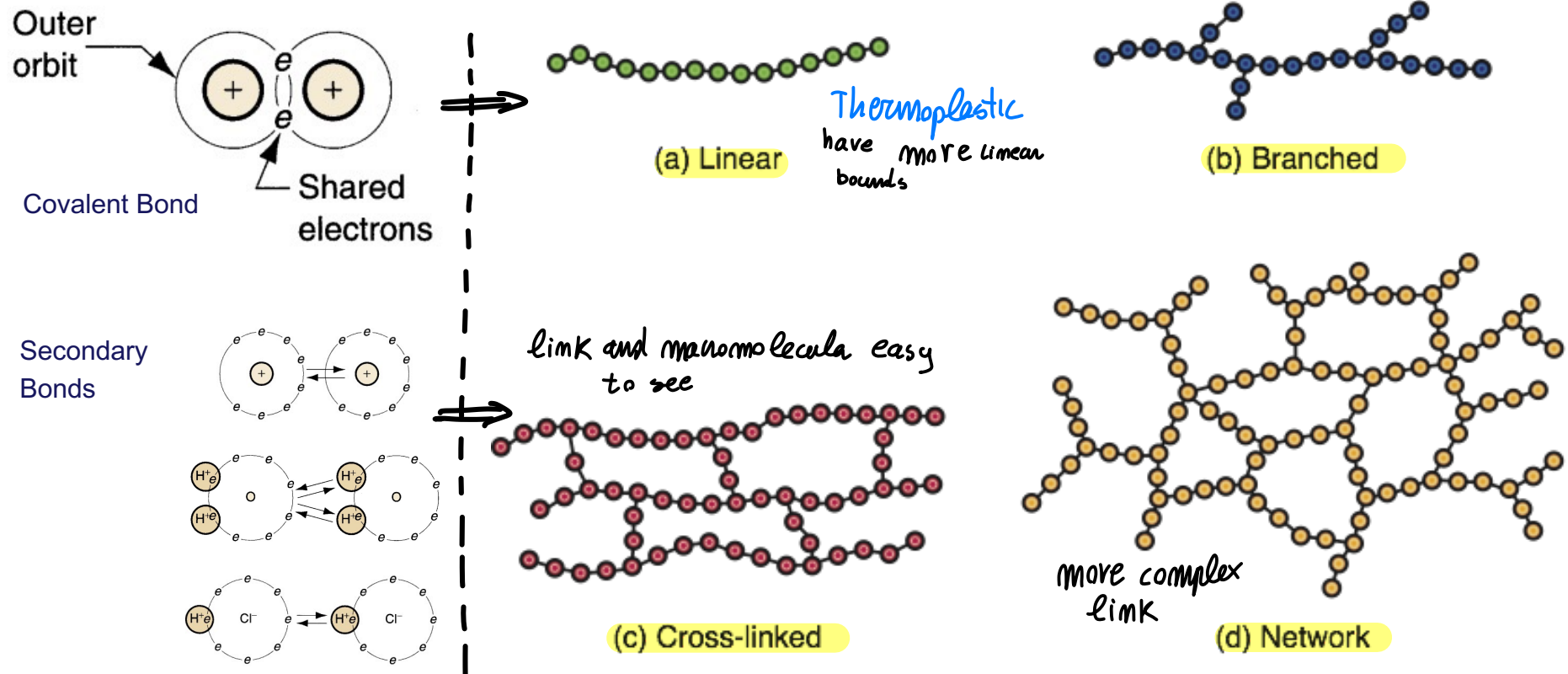


STRUCTURE OF POLYMERS

MACROMOLECULA : covalent
bond based along different elements

49

During polymerization, the monomers are linked together by covalent bonds (a primary bond), forming a polymer chain. The polymer chains are, in turn, held together by secondary bonds, such as van der Waals bonds, hydrogen bonds, and ionic bonds. *↑ More level of bonds in the material*
(micromolecule stay together thanks to 2ND bonds)



PRIMARY: create a basic linear shape (a)
or more complex ones

(b) Branches: more bonds in some position
PRIMARY

(c) links along macromolecule!
CROSS-LINK among

(d) Network $\rightarrow$ is like a NET, complex link!

im. thermoplastic polymers { linear
Branched

• thermoset polymers { cross-link $\Rightarrow$ NON-REVERSABLE!
Network

structure related to some kind of process

▶ process creating CROSS-LINK is called CURING of
thermoset material
 $\Downarrow$
product as an unique macromolecule,
complex chain!

▶ linear macromolecular product $\Rightarrow$ linked by 2nd Bond
(easy to separate $\rightarrow$ remelt them)



POLYMER CRYSTALLINITY

CRISTAL structure on space

50

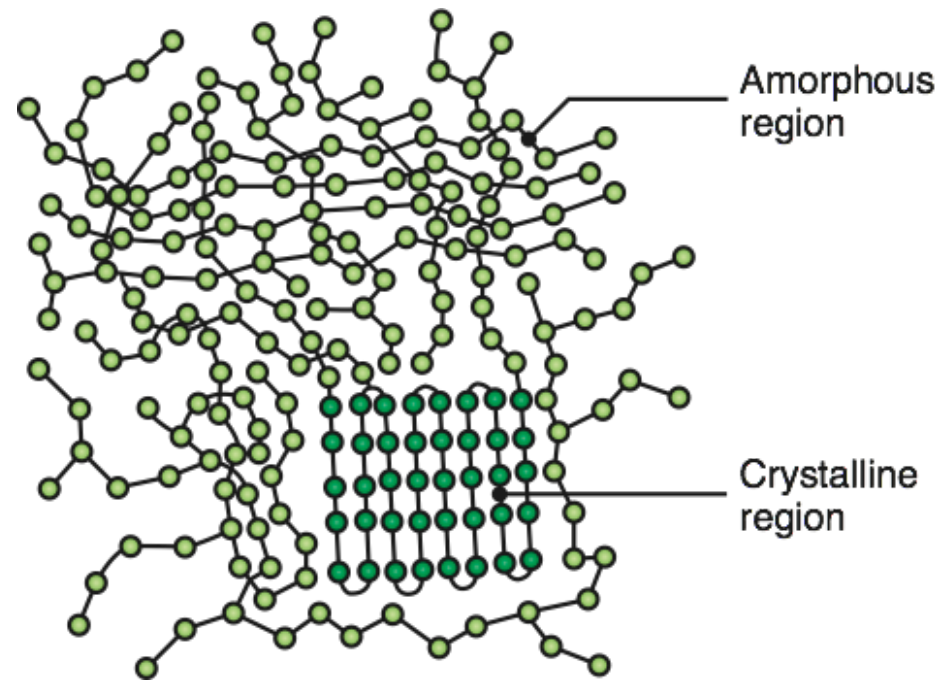
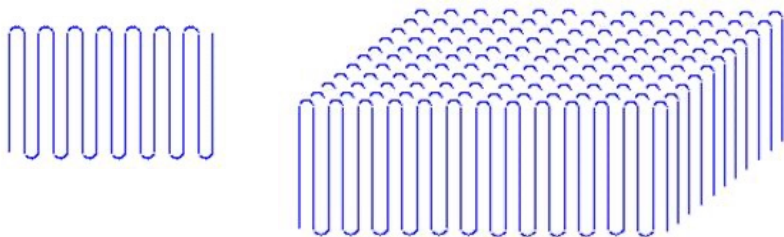
NOT REGULAR

Polymers are generally **amorphous**. The amorphous arrangement of polymer chains often is described as being like a bowl of spaghetti, all intertwined with each other.

In some polymers, however, it is possible to impart some **crystallinity** and thereby modify their characteristics. This arrangement may be fostered either during the synthesis of the polymer or by deformation during its subsequent processing.

usual of thermoplastic... REGULAR SHAPE

Degree of crystallinity: ratio between the weight of crystalline regions and the overall weight of the polymer (usually < 60 - 70%).



Those macromolecular can be folded on space with a

similar crystal structure

↓
CRYSTALLINE similar also obtained on polymers

CRYSTALLINITY ~~NO~~ in thermoset! Random structural

↓

thermoplastic instead
has a degree of crystallinity

usually amorphous
(NOT REGULAR)

↓

material with regular shape

↳ essential
for melting
behav.



TRANSITION TEMPERATURES OF POLYMERS

100% crystalline polymer: sharp liquid-solid transition temperature

(melting temperature - T_m).

Melting of MATERIALS... SOLID to LIQUID
in polymers (VISCOUS LIQUID) → sudden change

Amorphous polymers do not have a specific melting point, they undergo a distinct change in their mechanical behaviour across a narrow range of temperatures.

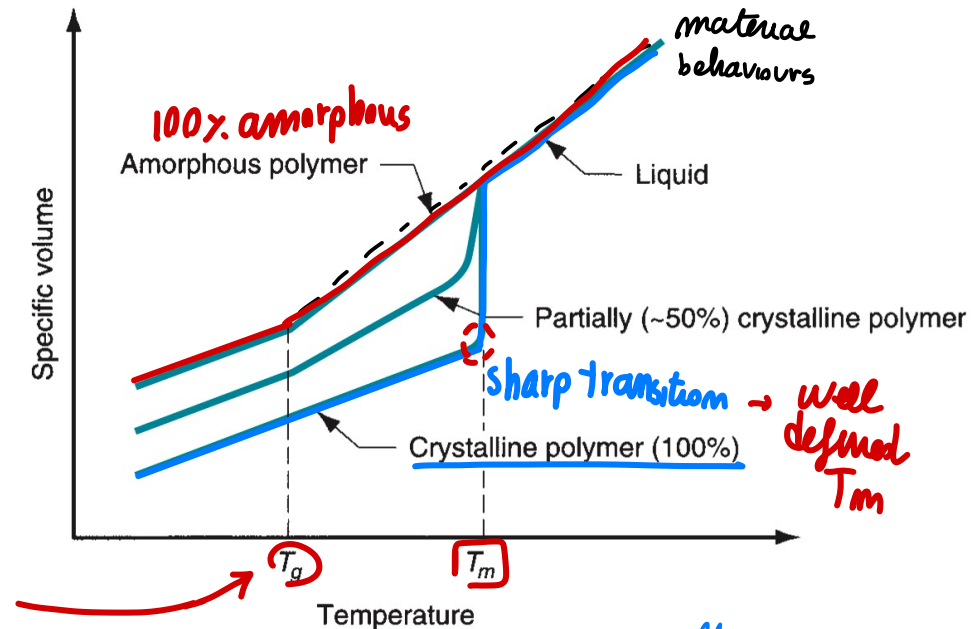
At low temperatures, they are hard, rigid, brittle, and glassy; at high temperatures, they are rubbery or leathery.

The temperature at which a transition occurs is called the glass-transition temperature (T_g), also called the glass point or glass temperature.

if we can have 100% CRYSTALLINE of thermoplastic and HEAT it we see a MELTING POINT

SOLID → • → LIQUID

(VISCOUS LIQUID)



IF 100% crystalline... $\Rightarrow$ softem behaviour with temperature
(high viscosity liquid)

Th: go down to Glass transition temperature T_g
 $\downarrow$
brittle materials (amorphous)
(Glassie behav.)

100% Amorphous $\Rightarrow T_g \Rightarrow$ if $T < T_g$ GLASS LIKE: britleness

if $T > T_g \Rightarrow$ I'll be high viscosity
shapeable material



if CRYSTALLINE % and Amorphous $\Rightarrow$ both behaviours...
GLASS transition / MELTING of crystalline parts



TRANSITION TEMPERATURES OF POLYMERS

CHARACTERIZE
material
through
this temp.

Glass-transition and Melting Temperatures of Some Polymers

Material	T_g (°C)	T_m (°C)
Nylon 6,6	57	265
Polycarbonate	150	265
Polyester	73	265
Polyethylene		
High density	-90	137
Low density	-110	115
Polymethylmethacrylate	105	—
Polypropylene	-14	176
Polystyrene	100	239
Polytetrafluoroethylene	-90	327
Polyvinyl chloride	87	212
Rubber	-73	—

← degree of
CRYSTALLINITY
portion of
material
CRYSTAL,
regular in
space!

if 100% AMORPHOUS
(only T_g)



THERMOPLASTIC POLYMERS (TP)

Thermoplastic polymers can be heated from solid state to viscous liquid and then cooled back down to solid:

- Heating and cooling can be repeated many times without degrading the polymer;
- Reason: TP polymers consist of linear and/or branched macromolecules that do not cross-link.

Thermosets and elastomers change chemically when heated, which cross-links their molecules and permanently cures these polymers.



THERMOSETTING POLYMERS (TS)

TS polymers are distinguished by their highly cross-linked three-dimensional, covalently-bonded structure.

Chemical reactions associated with cross-linking are called curing or setting.

In effect, formed part becomes a large macromolecule.

Always amorphous and exhibits no glass transition temperature.

Cannot be re-melted, instead they degrade, burn, or char.

↳ NOT Recyclable ! 😞



CROSS-LINKING (CURING) OF TS POLYMERS



Three categories:

- ① Temperature-activated systems: curing caused by heating;
- ② Catalyst-activated systems: curing by small amounts of a catalyst added to the liquid polymer;
- ③ Mixing-activated systems: two starting ingredients react to cause curing.

Curing is done at the fabrication plants that make the parts rather than the supplier of the starting materials.



Polymers capable of large elastic deformation when subjected to relatively low stresses:

- Some can be extended 500% or more and still return to their original shape.
- Two categories:
 1. Natural rubber - derived from biological plants;
 2. Synthetic polymers - produced by polymerization processes like those used for TP and TS polymers.

⇒ elongation properties!



VULCANIZATION

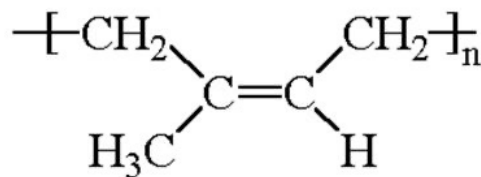
processing of adding new chemical element to create **CROSS-LINK**
typical of **TIRES** production

↓ **CROSS-LINKING** needed for elongation

Vulcanization is the term for curing (cross-linking) in the context of natural rubber (and certain synthetic rubbers).

Typical cross-linking in rubber is one to ten links per hundred carbon atoms in the linear polymer chain, depending on degree of stiffness desired.

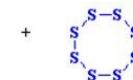
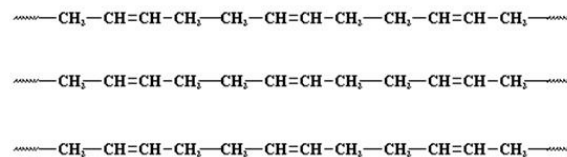
Considerable less than cross-linking in thermosets.



polyisoprene

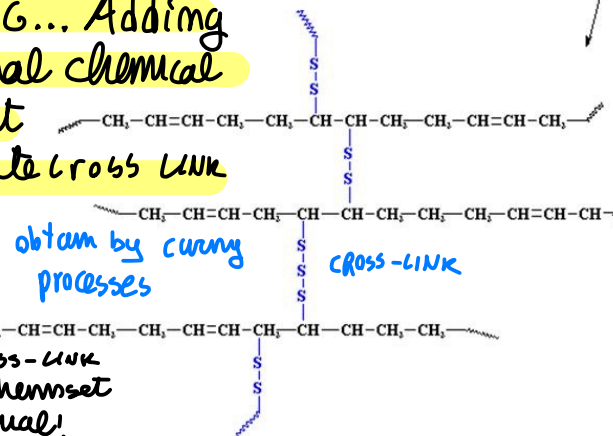
START from a long macromolecule

to enhance elastic behavior.



Sulfur or sulphur

process of **VULCANIZ.**
CURING... Adding additional chemical element to create **CROSS LINK**



obtained by curing processes

less **CROSS-LINK** than thermoset material!

CROSS-LINK creation
huge elongation admitted, even if loading mechanically
↓
HIGH ELONGATION > thermoplastic but reversible (elastic recovery)



VULCANIZATION

58

material
without cross link

With some
cross link



apply LOAD



Remove LOAD



different
situation



different
behaviours in
the two cases



After load
came back
as previously!

Elastomer
without cross-linking

Elastomer
with cross-linking



different kind of elements!
different usage

General Recommendations for Plastic Products

Design requirement	Typical applications	Plastics
Mechanical strength	Gears, cams, rolls, valves, fan blades, impellers, pistons	Acetals, nylon, phenolics, polycarbonates, polyesters, polypropylenes, epoxies, polyimides
Wear resistance	Gears, wear strips and liners, bearings, bushings, roller blades	Acetals, nylon, phenolics, polyimides, polyurethane, ultrahigh-molecular-weight polyethylene
Frictional properties		
High	Tires, nonskid surfaces, footwear, flooring	Elastomers, rubbers
Low	Sliding surfaces, artificial joints	Fluorocarbons, polyesters, polyimides
Electrical resistance	All types of electrical components and equipment, appliances, electrical fixtures	Polymethylmethacrylate, ABS, fluorocarbons, nylon, polycarbonate, polyester, polypropylenes, ureas, phenolics, silicones, rubbers
Chemical resistance	Containers for chemicals, laboratory equipment, components for chemical industry, food and beverage containers	Acetals, ABS, epoxies, polymethylmethacrylate, fluorocarbons, nylon, polycarbonate, polyester, polypropylene, ureas, silicones
Heat resistance	Appliances, cookware, electrical components	Fluorocarbons, polyimides, silicones, acetals, polysulfones, phenolics, epoxies
Functional and decorative	Handles, knobs, camera and battery cases, trim moldings, pipe fittings	ABS, acrylics, cellulotics, phenolics, polyethylenes, polypropylenes, polystyrenes, polyvinyl chloride
Functional and transparent	Lenses, goggles, safety glazing, signs, food-processing equipment, laboratory hardware	Acrylics, polycarbonates, polystyrenes, polysulfones
Housings and hollow shapes	Power tools, housings, sport helmets, telephone cases	ABS, cellulotics, phenolics, polycarbonates, polyethylenes, polypropylene, polystyrenes